

Text Classification on Social Media

NLP Assignment 1

Developed by: António Romão, Guilherme Gonçalves, Pedro Leitão

Data Provenance & Annotation

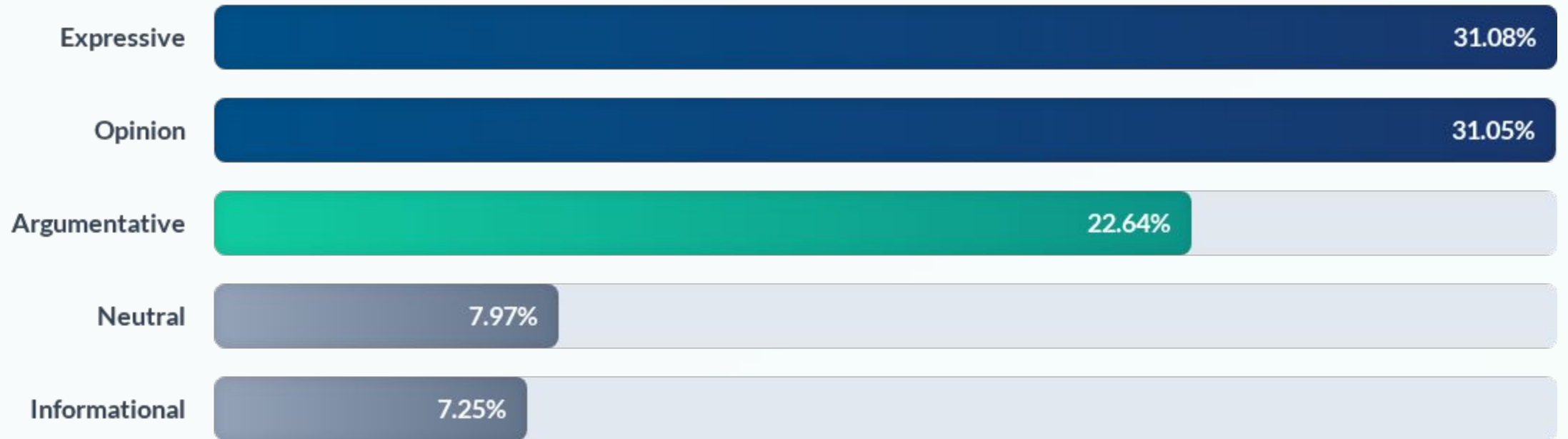
Origin & Collection

- **Platforms:** YouTube, Hacker News, MetaFilter, Reddit, and BlueSky.
- **Timeline:** Sourced between 2024 to mid-February 2026.
- **Scope:** Filtered via 'politics' or 'US Politics' search queries.
- **Quality Control:** Restricted to posts containing between 10 and 300 comments to prevent dominance by highly active threads.

Annotation Methodology

- **Nuanced Labels:** Argumentative, Informational, Opinion, Expressive, and Neutral.
- **Human Seeding:** 100 comments manually annotated and reconciled to resolve definition boundaries.
- **LLM Ensemble:** ~70k comments labeled at scale via Batch API using Gemini Flash 3, ChatGPT 5.1, and Claude Haiku 4.5.
- **Consensus:** Ground truth established via a strict 2-of-3 majority vote.

EDA: Class Distribution & Text Length



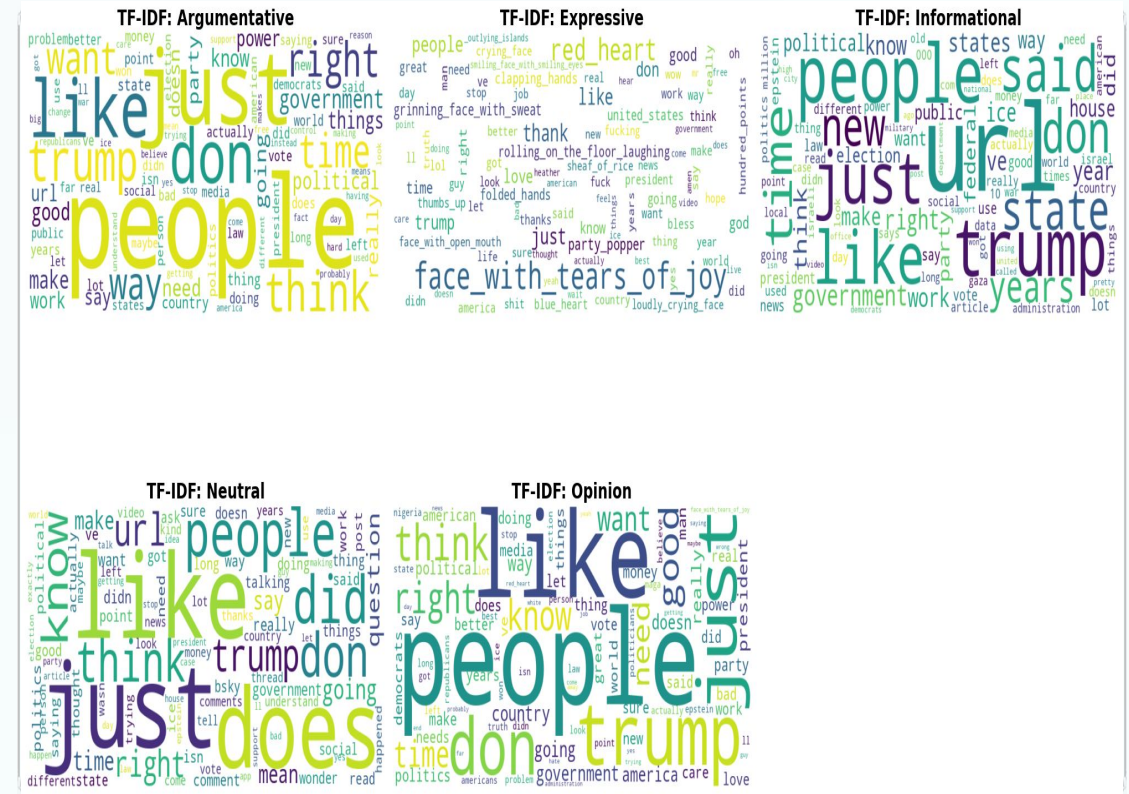
Observation: Significant class imbalance exists, driven by emotional and opinionated discourse. Comment lengths are heavily right-skewed (median 17 words, max 2162). Structure varies heavily by intent: **Argumentative** comments have a median length of 51 words, compared to just 10 words for **Expressive** reactions.

Lexical & Artifact Signatures

Applying Term Frequency-Inverse Document Frequency (TF-IDF) revealed distinct vocabulary signatures perfectly aligned with our dataset definitions:

- **Expressive:** Overwhelmingly dominated by repetitive emoji text strings (e.g., *face_with_tears_of_joy*) and strong reactive language.
- **Informational:** Identified strongly by the *[url]* token, alongside factual reporting verbs ("said") and objective nouns.
- **Opinion vs Argumentative:** "Opinion" revolves around short value judgments ("bad", "smart"), while "Argumentative" relies on length, reasoning connectors, and epistemic verbs.

These unique signals justify preserving emojis and masked URLs as explicit features.



Targeted Pre-processing Pipeline



Normalization

Enforced strict lowercasing and utilized contraction expansion (e.g., converting "don't" to "do not"). This explicitly preserved negation markers, which are critical for differentiating opinions from counter-arguments.



Punctuation

Applied selective regex removal, replacing non-alphanumeric characters with spaces. This flawlessly preserved engineered structural tags ([url]) and emojis (👉), enabling safe whitespace tokenization.



Lemmatization

Consolidated inflected forms via POS-tagged Lemmatization instead of brute-force stemming. This prevented the destruction of semantic nuance while continuing to bypass and protect our engineered artifacts.

Feature Representation Strategy

Sparse & Dense Matrices

- **Sparse (TF-IDF):** Configured with unigrams and bigrams, min_df=5, and max_df=0.90 to prune rare typos and ubiquitous fillers. Yielded an optimized 26,512 dimension space.
- **Dense (GloVe):** Deployed GloVe Twitter 100d with a Continuous Bag of Words (CBOW) average. Discovered that dense embeddings dropped OOV tokens like URLs and emojis.

Integration of Meta-Features

- **The Meta-Signals:** Structural differences are highly predictive. We extracted 5 meta-features: *word_count*, *char_count*, *emoji_count*, *contains_url*, *has_foreign_script*.
- **Concatenation:** Applied a Standard Scaler and appended these directly to both Sparse and Dense matrices. This ensured traditional algorithms retained the vital structural intent lost by GloVe.

Experimental Results: Model Selection

Model	Feature Space	Validation Accuracy	Validation Macro-F1
Baseline: Majority Class	-	0.3085	0.0943
Baseline: Multinomial NB	TF-IDF	0.5432	0.3772
Linear SVC	TF-IDF + Meta	0.5733	0.5194
MLP-128	GloVe + Meta	0.6268	0.5242
Tuned Logistic Regression	TF-IDF + Meta	0.5946	0.5574

Sparse TF-IDF representations consistently outperformed Dense GloVe embeddings. The Tuned Logistic Regression ($C=1.0$, `class_weight=balanced`) was selected as the final model, soundly beating all baseline floors.

Test-Set Evaluation (Final Tuned Model)

Class	Precision	Recall	F1-Score	Support
Neutral	0.2468	0.4698	0.3236	779
Opinion	0.5950	0.5450	0.5689	3257
Argumentative	0.7049	0.6744	0.6893	2423
Expressive	0.7487	0.6251	0.6814	3289
Informational	0.4202	0.5333	0.4701	810

Test Accuracy: 0.5932 | Test Macro-F1: 0.5467

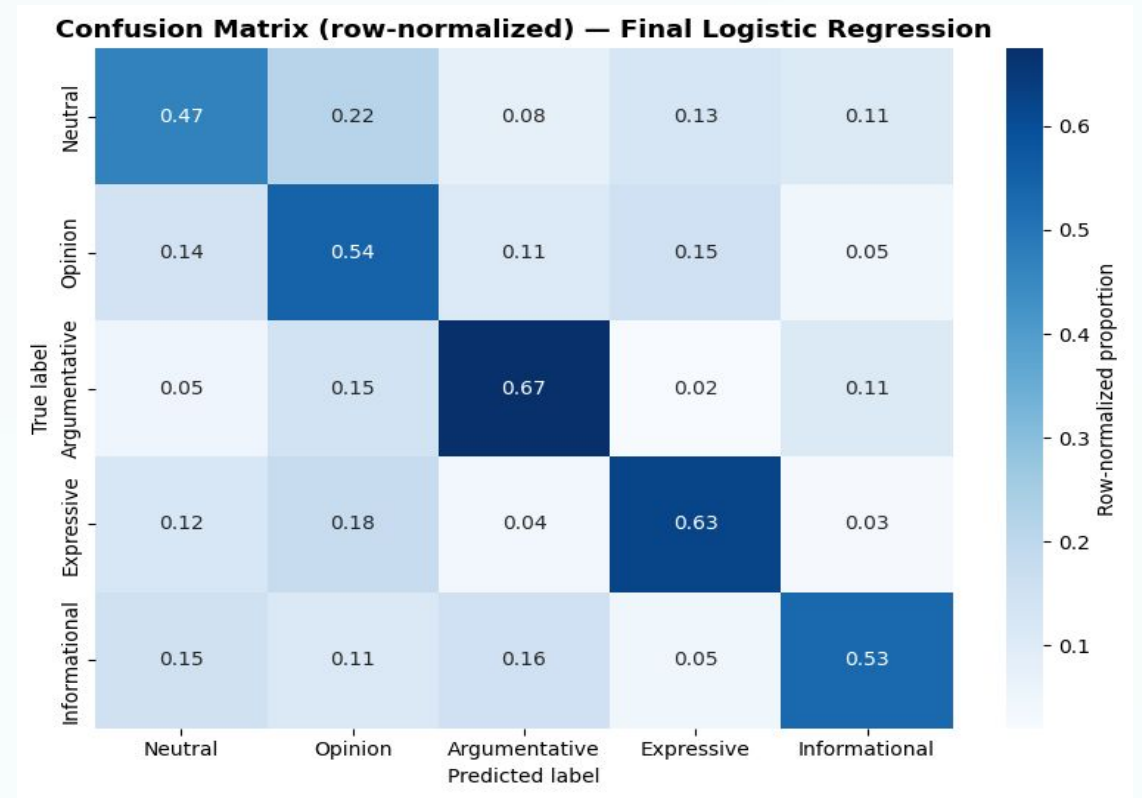
Argumentative performs remarkably well despite being a minority class, successfully captured by specific TF-IDF n-grams and length meta-features. Neutral struggles due to its nature as a noisy "catch-all" category.

Error Analysis: Analyzing the Confusion

The top model confusions occur across the subjective continuum:

Expressive vs. Opinion and *Opinion vs. Argumentative*.

- **LLM Label Ambiguity:** The "ground truth" reflects a 2-of-3 LLM majority. Many comments intrinsically mix intents (e.g., starting neutral, ending with an emotional opinion).
- **Foreign-Language Failures:** Dense vectors collapsed to zero on non-English text because they lacked English vocabulary markers.
- **Implicit Stance & Sarcasm:** Identifying sarcasm relies on real-world context and irony, completely bypassing surface-level bag-of-words architectures.



Project Conclusions

Classical Machine Learning combined with rigorous feature engineering (TF-IDF + engineered structural meta-features) extracts massive textual signal, successfully closing **40% of the gap** (19% improvement) to state-of-the-art Transformer models (like BERTweet) without deep learning.

Sarcasm, mixed-intents, and annotation subjectivity remain the hard upper bounds for traditional bag-of-words classifiers.

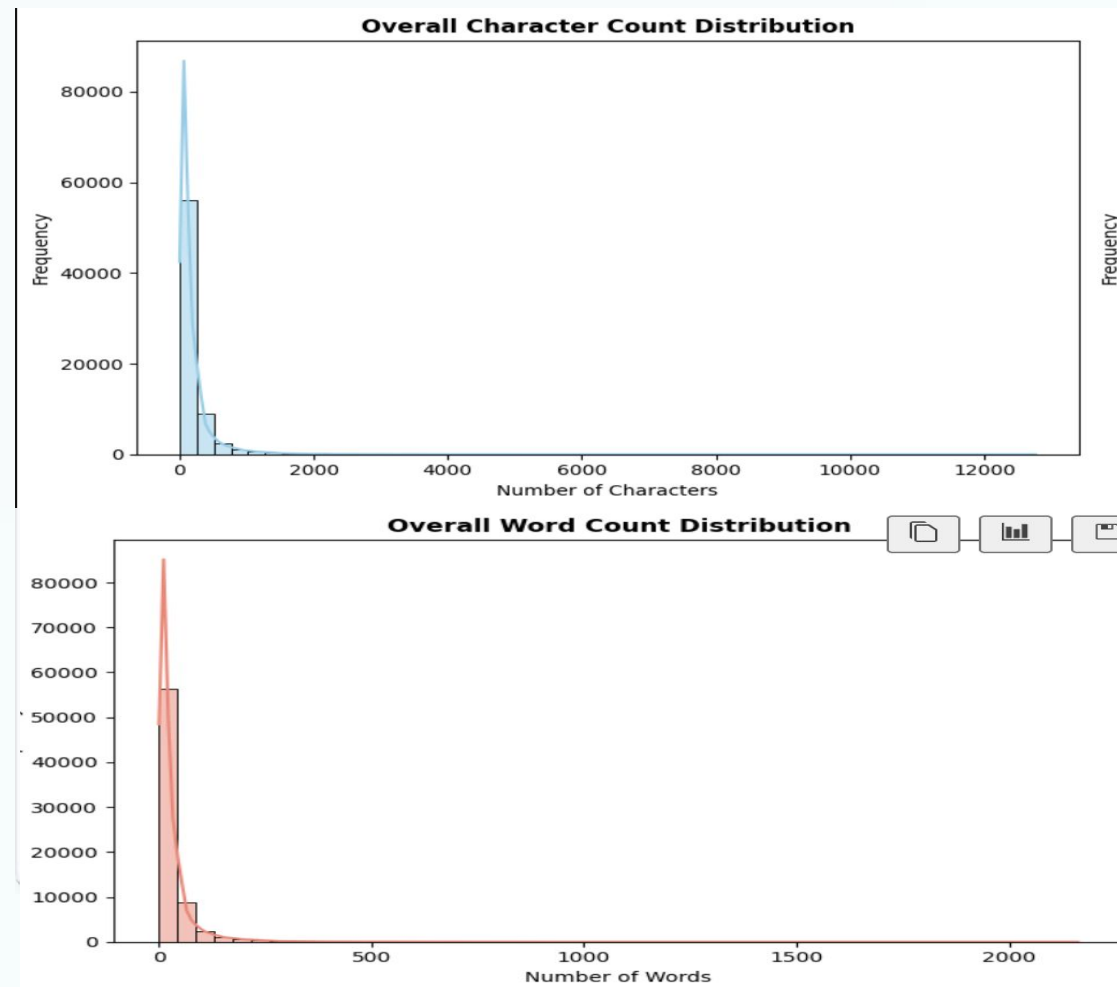
Appendix & Technical Details

Backup documentation containing in-depth analyses, dataset statistics, feature weighting logs, and related work comparisons.

Dataset Statistics: Text Length Outliers

The overall character and word count distributions are extremely right-skewed, prompting strict limits on the TF-IDF feature space.

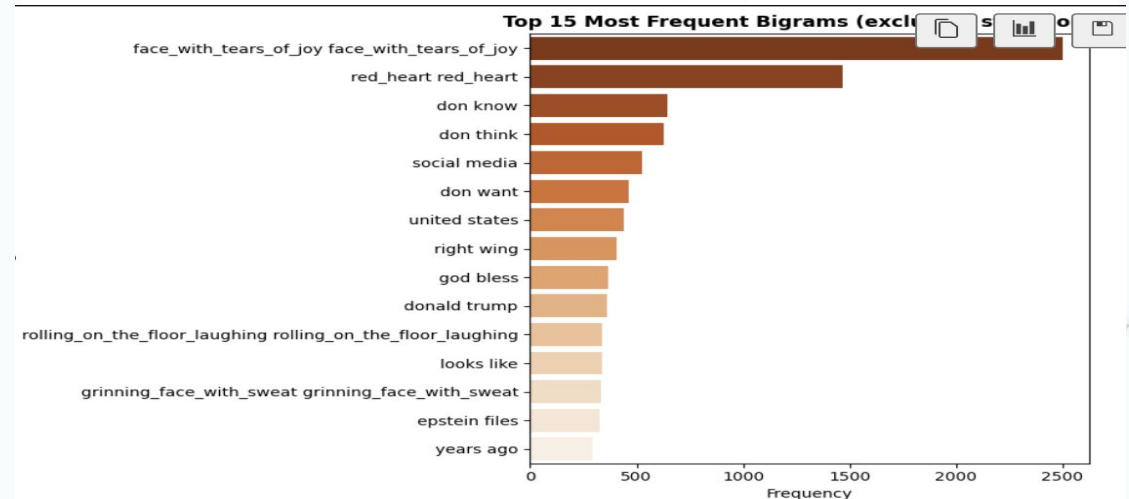
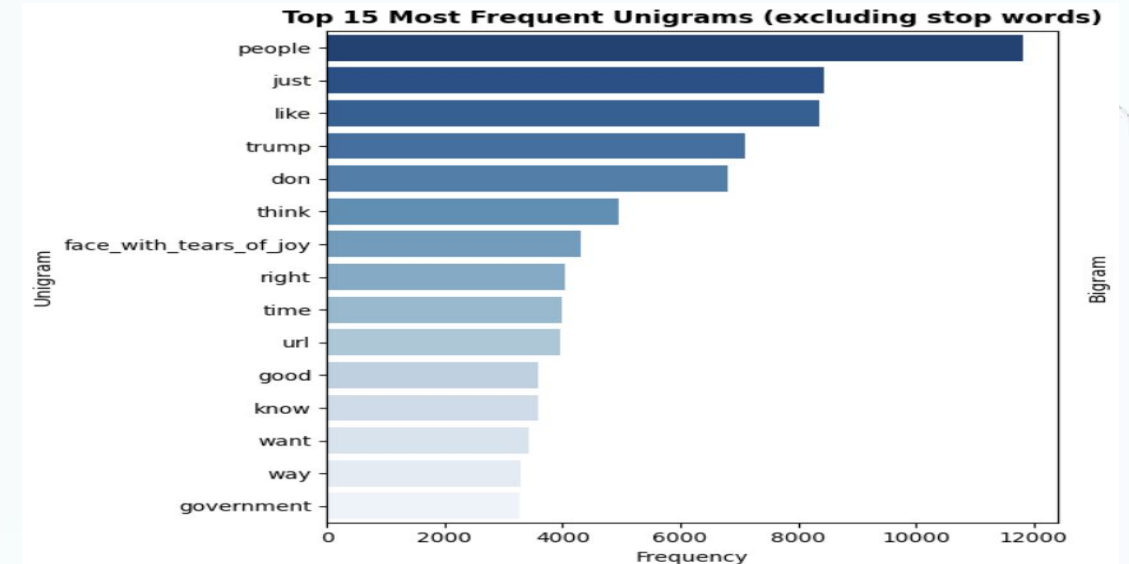
- **Character Counts:** Mean of 193.7 chars, but extends to a maximum outlier of **12,768 characters**.
- **Word Counts:** Median is just 17 words, while the maximum is **2,162 words**.
- **Pre-processing Impact:** Because extreme rants contain idiosyncratic vocabulary, ``min_df=5`` and ``max_df=0.90`` parameters were strictly required to prevent the sparse matrix from expanding exponentially.



Lexical Baseline: Raw N-Gram Frequencies

Before applying TF-IDF, a simple count vectorization analysis highlighted the dominance of conversational fillers and artifacts.

- **Conversational Junk:** Top unigrams included *"just"*, *"like"*, and *"people"*.
- **Engineered Tokens:** Repeated emojis (e.g., *face_with_tears_of_joy* twice) formed the most frequent bigram by a massive margin (2501 occurrences).
- **Tokenization Errors:** The presence of *"don think"* and *"don know"* validated our custom contraction expansion step (fixing fragmented *"don't"* tokens).



Pre-Processing: Stemming vs. Lemmatization

Aggressive Stemming (Rejected)

Raw stemming aggressively chops word endings, destroying semantic clarity and creating non-words. This limits interpretability during error analysis.

```
['read', 'comprehens', 'good',  
'thing', 'go', 'back', 'tri',  
'number', 'talk', 'not',  
'agre', 'equal', 'anim',  
'equa', 'othersth']
```

POS-Tagged Lemmatization (Adopted)

Reduces inflections to proper dictionary lemmas using context, preserving meaning while safely bypassing engineered tags (e.g., [url]).

```
['read', 'comprehension',  
'good', 'thing', 'go',  
'back', 'try', 'number',  
'talk', 'not', 'agree',  
'equally', 'animal', 'equal']
```

LogReg Hyperparameter Grid Search

Regularization (C)	Class Weight	Validation Macro-F1
1.0	balanced	0.5574
0.5	balanced	0.5570
2.0	balanced	0.5474
0.1	balanced	0.5407
2.0	None	0.5360
0.1	None	0.4703

The grid search evaluated variations around the scikit-learn default (C=1.0). The `class_weight='balanced'` parameter proved absolutely critical for lifting Macro-F1 by properly weighting the minority classes (Neutral, Argumentative, Informational).

Learned Feature Weights (Part 1)

Argumentative Class

Model relies on structural volume and propositions, avoiding emotional pleasantries.

- **+3.14** char_count
- **+3.00** would
- **+2.85** not (Protected negation)
- **+2.50** problem
- **-2.59** thank
- **-1.89** word_count

Opinion Class

Dominated by value judgments and absolutes, actively penalized by informational markers.

- **+2.36** need
- **+2.23** corrupt
- **+1.96** bad
- **+1.94** always
- **-1.88** [url]
- **-1.83** char_count

Learned Feature Weights (Part 2)

Expressive Class

Perfectly captures short, venting reactions. Penalized by concrete political terminology.

- **+6.30** thank
- **+4.83** f*ck
- **+4.82** face_with_tears_of_joy
- **+2.78** wow
- **-2.57** election
- **-2.36** char_count

Informational Class

Driven entirely by external references and explicit entities. Suppresses cognitive verbs.

- **+3.91** www
- **+3.38** char_count
- **+2.91** [url]
- **+2.63** encrypt
- **-3.79** need
- **-3.40** think

Interpreting the "Neutral" Class



Positive Interrogatives

The model accurately maps this "catch-all" class to specific meta-commentary and rhetorical question vocabulary.

- **+3.82** wonder
- **+3.58** question
- **+2.79** curious
- **+2.36** elaborate



Negative Penalties

To identify Neutrality, the model actively avoids highly charged political entities, strong sentiment, and long texts.

- **-2.59** trump
- **-2.33** char_count
- **-2.11** evil / good
- **-1.99** hate

Qualitative Error Analysis



LLM Label Ambiguity (True: Expressive | Pred: Opinion) "shut it down. we won't even notice."

Analysis: The gold label is derived from a 2-of-3 LLM majority. This comment reads distinctly as an opinion, demonstrating fuzzy label boundaries where human or model annotators might reasonably disagree.



Foreign-Language Failures (True: Opinion | Pred: Neutral) "ну просто русский президент с душой"

Analysis: Multilingual text collapses to a zero-vector because these tokens don't exist in the English GloVe vocabulary and are stripped by the min_df parameter, forcing the model into a blind fallback prediction.



Sarcasm & Mixed Intent (True: Expressive | Pred: Opinion) "remember that time he gave himself a tbi on national television"

Analysis: Surface-level bag-of-words models cannot detect sarcasm or implicit reference, as the irony is carried by contextual world-knowledge rather than isolated lexical cues.

Related Work: Head-to-Head Comparison

Model	Architecture Type	Parameters	Validation Macro-F1
Majority-class baseline	Trivial	-	0.0943
Multinomial NB	Classical (TF-IDF)	-	0.3772
Our Tuned LogReg	Classical (TF-IDF + Meta)	-	0.5574
experiment_labels_bert_base	Transformer (Fine-tuned)	~110M	0.7295
BERTweet-large-self-labeling	Transformer (Domain-specific)	~355M	0.7817

Conclusion: Our classical NLP pipeline establishes a robust baseline previously missing from the dataset literature. Through rigorous feature engineering, we successfully closed over **40% of the gap** between the simple Naive Bayes baseline and state-of-the-art 355M parameter Transformers, reaching the maximum capacity of non-contextual architectures.